

Q: Why does increasing the number of bits not necessarily improve accuracy?

AI Model: 1

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A:

Increasing the number of bits in an Analog-to-Digital Converter (ADC) can actually degrade performance due to quantization noise, a fundamental limitation inherent to ADCs.

Quantization noise arises from the fact that the ADC's resolution is limited by its finite number of output levels (2^n for n -bit resolution). When representing a continuous-time analog signal with a discrete-time digital representation, this quantization introduces an error proportional to the difference between the actual and quantized values.

Mathematically, this can be expressed as: $\epsilon_{quant} = V_{ref}/2^{n+1}$, where V_{ref} is the ADC's reference voltage, and n is the number of bits. As the number of bits increases, the magnitude of quantization noise actually decreases due to the increased resolution. However, this improvement is overshadowed by another factor: increasing the sampling rate required to maintain a given Signal-to-Quantization Noise Ratio (SQNR). In practice, as the number of bits increases, so does the power consumption and area of the ADC, making it less efficient and more complex. Thus, while additional bits can provide higher resolution, they come at the cost of decreased SQNR due to increased quantization noise, which ultimately limits accuracy..

AI Model: 2

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A:

In theory, increasing the number of bits (N) improves Signal-to-Quantization Noise Ratio (SQNR) by $6.02 + 1.2(N \text{ [dB]})$ because quantization noise power scales as $\frac{V_{ref}^2}{12 \cdot LSB^2} = \frac{V_{ref}^2}{12 \cdot (\frac{V_{FS}}{2^N})^2}$.

However, in practice, accuracy improvement stalls due to non-idealities that scale with speed and resolution. These include static errors like comparator mismatch (σ_{offset}) and

limited intrinsic linearity (INL/DNL deviation), which become dominant when the theoretical step size (LSB) approaches or falls below their RMS error floor; for example, the effective number of bits (ENOB) is often limited by $ENOB \approx \log_2(1/\sigma_{\text{offset}})$. Dynamic errors are critical at high speeds: sampling jitter (τ_{rms}) degrades the effective voltage bandwidth according to $V_{\text{error}}/V_{\text{in}} \propto 2\pi f_{\text{out}} \cdot \tau_{rms}/SNR$, effectively capping the achievable Signal-to-Noise Ratio (SNR) regardless of quantization resolution. Furthermore, finite gain/bandwidth (opamp A_v , BW) limits the accuracy of charge sharing and differential stages, meaning that exceeding a certain operational speed or resolving power threshold introduces systematic circuit nonlinearities that swamp the incremental improvement from added bits..

AI Model: 3

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Increasing bit resolution *expands dynamic range*—the ratio between full-scale input (FSR) and the smallest detectable change—by **$2\times$ per bit** (e.g., 16-bit spans ± 0.3 ppm of FSR, while 8-bit spans ± 4 ppm), but it does not inherently improve *accuracy* because accuracy depends on both resolution and error sources that scale with resolution. For example, **integral nonlinearity (INL)** errors (e.g., comparator offset or DAC mismatch) grow in absolute terms as bit count increases, often dominating precision beyond ~ 12 – 14 bits unless carefully mitigated (e.g., via calibration). Similarly, **thermal noise** in the input path scales with resolution (kT/C noise in capacitors or op-amp input-referred noise), requiring larger sampling capacitors or lower gain stages to avoid overwhelming the LSB. **Clock jitter** introduces time-domain errors that manifest as phase noise in $\Delta\Sigma$ ADCs or spurious tones in SAR/pipeline ADCs, whose impact worsens with higher oversampling ratios (common in high-resolution designs). Finally, **reference voltage stability** and **power supply ripple** introduce fixed offsets that become proportionally larger relative to the LSB as resolution increases. Thus, while 24-bit ADCs can achieve sub-ppm accuracy in ideal conditions, real-world implementations often plateau at ~ 16 – 20 bits due to these scaling limitations—unless active error correction (e.g., multi-step calibration) is employed..

AI Model: 4

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A:

Increasing ADC resolution (N) reduces quantization step size $\Delta V = V_{REF}/2^N$, yet accuracy is fundamentally bounded by total error variance rather than nominal bit count; if systematic errors such as INL, offset, reference drift, or thermal noise exceed one LSB, adding higher-order codes merely partitions these fixed disturbances into additional bins without improving true signal fidelity. This behavior manifests in the Effective Number Of Bits ($ENOB$), where $SNDR \approx 6.02(ENOB) - 174$ dB dictates that performance saturates once a system error floor—driven by op-amp finite gain, clock jitter σ_j , or component mismatch—is larger than the quantization noise of subsequent bits ($E_{sys} > \Delta V'_{LSB}$). Consequently, blindly scaling bit-counts without addressing power consumption, signal bandwidth settling, and voltage reference stability results in diminishing returns where the ADC cannot resolve signals beyond its intrinsic error floor, causing $ENOB$ to remain significantly lower than nominal resolution..
